

Yifan Li

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Education

University of Oxford, MSc in Mathematics and Foundations of Computer Science Oct 2025 - Sept 2026

- **Selected Mini Projects:** Algebraic Topology (Configuration Spaces), Computational Algebraic Topology (Applications of Discrete Morse Theory), Geometric Group Theory (Quasi-convex Subgroups), Infinite Groups (Wreath Products), Low-dimensional Topology & Knot Theory (Unknotting Numbers)

Xi'an Jiaotong-Liverpool University (XJTLU), BS in Applied Mathematics Sept 2021 - July 2025

- **Part of Coursework:** Topology, Geometry of Curves and Surfaces, Algebra, Discrete Mathematics, Dynamic Modelling, Functional analysis, PED, Relativity

Research Interests

Geometric group theory; mapping class group; $\text{Out}(F_n)$; space of nonpositive curvature; low-dimensional topology.

Publication

Packing Edge-colorings of Subcubic Outerplanar Graphs

Sijin Li, **Yifan Li**, Xujun Liu. Packing Edge-colorings of Subcubic Outerplanar Graphs, *Discrete Applied Mathematics* 391(15): 68-82, 2026. <https://doi.org/10.1016/j.dam.2026.04.033>

Selected Research Experiences

Dissertation: The Model Theory of Curve Graphs & $\text{Out}(F_n)$ Supervised by Prof. Dawid Kielak 04/2026 - 09/2026

Studied rigidity phenomena relating mapping class groups, outer automorphism groups of surface groups, and automorphism groups of curve complexes, from both topological and model-theoretic viewpoints.

- The project began by studying the relations among the above three groups in the surface topology viewpoint.
- Then review the paper *The model theory of the curve graph* and provide a detailed proof of the theorem from the model theory viewpoint.
- Extend the methodology to $\text{Out}(F_n)$ and show that geometric graphs corresponding to the embedded spheres give $\text{Out}(F_n)$ as the automorphism group iff such a geometric graph is bi-interpretable with the augmented Cayley graph of $\text{Out}(F_n)$.

Final Year Project: Exploration of Geometric Group Theory and Mapping Class Groups Supervised by Prof. Robert Tang 09/2024 - 06/2025

Learned the foundational aspects of geometric group theory, with a particular focus on understanding mapping class groups, and explored the hyperbolicity of curve graphs.

- Read Clara Loh's *Geometric Group Theory, An Introduction* to understand the basics of geometric group theory.
- Advanced to studying Farb and Margalit's *A Primer on Mapping Class Groups* and understanding mapping class groups.
- Studied the Gromov hyperbolicity and actions on hyperbolic spaces. Read *A note on acylindrical hyperbolicity of Mapping Class Groups* by Piotr and Alessandro to study the hyperbolicity of curve graph.

Validation of Packing Coloring Problem 06/2024 - 11/2024

Studied and validated the Packing Coloring conjecture proposed by Hocquard et al. regarding the packing coloring of subcubic outerplanar graphs, with a focus on showing the existence of $(1, 2^5)$ -packing edge-coloring and $(1^2, 2^3)$ -packing edge-coloring

- Proved packing edge-colouring results for subcubic outerplanar graphs using minimal-counterexample and structural arguments.
- Established sharpness examples and bounds for several generalized packing-colouring parameters.
- Resulted in a publication in *Discrete Applied Mathematics* (2026).

Selected Seminars and Study Groups

- **Geometry of Nonpositive Curvature:** Reading chapters III.H and Γ from the book *Metric Spaces of Non-Positive Curvature* by Martin Bridson and André Haefliger. 10/2026 - 12/2026
- **Bass-Serre Theory:** Reading the book *Trees* by Jean-Pierre Serre. 10/2026 - 12/2026
- **3-Manifolds:** Reading the book *An Introduction to Geometric Topology* by Bruno Martelli. 10/2026 - 12/2026

Awards and Honors

Best Overall Academic Performance	07/2025
Best Performance in Final Year Project	07/2025
The Rajendra Bhansali Prize (Top 1%)	07/2025
University Academic Excellence Award (Top 5%), XJTLU	08/2022, 08/2023, 08/2024
Jiangsu Province Outstanding Students Award	08/2024

Conference and Summer School Attended

- Summer School on Geometric Group Theory, BICMR** 06/2025
Geometric Group Theory and Related Topics, BICMR
- Topics included median spaces, $CAT(0)$ cube complex and ultrafilters
 - Get a brief overview of HHS with intuitions
 - Learned about Garside shadows and cone parts together with Shi components in Coxeter groups

Volunteer and Community Experience

Vice Minister of Academic Department Peer Tutoring Club	05/2022 - 05/2023
Volunteer GAOKAO Consultation Fair	07/2022, 07/2025